AI RAG Waterproofing Chatbot

Simple Project Overview

A practical AI assistant that helps users understand waterproofing problems, find suitable construction-material solutions, and generate useful technical outputs.

1. General Idea

The project is a chatbot specialized in waterproofing and construction materials. It uses Retrieval-Augmented Generation (RAG) to search a knowledge base of trusted documents and generate answers based on retrieved information. Users can describe a problem or ask a technical question and receive guidance with supporting sources.

2. Purpose

The goal is to make technical information easier to access and help users compare possible solutions. The chatbot can explain products and application methods, recommend options based on available information, and create practical outputs instead of only giving text answers.

3. Main Features

Feature	What it does
Question answering	Answers questions using trusted technical documents and provides source references.
Solution recommendations	Suggests possible waterproofing systems based on details such as surface, location, and type of water infiltration.
Technical diagrams	Generates simple diagrams or layer cross-sections to explain how a proposed waterproofing system is arranged. These are illustrative visuals, not certified construction drawings.
Product comparison report	Creates a report comparing selected products by properties, suitable applications, advantages, limitations, and source references when available.

4. Dataset Sources

The knowledge base can include local PDF documents, product catalogues, technical data sheets (TDS), safety data sheets (SDS), installation guides, standards or technical guidance that the project is allowed to use, and official manufacturer websites. Product comparisons will only be as reliable as the available data; missing values should be marked rather than guessed.

5. Main Challenges

• Data quality: documents may be incomplete, outdated, or difficult to extract.

• Website collection: scraping must respect website terms, copyright, and access restrictions.

• Retrieval accuracy: the chatbot must find the right passages for each question.

• Recommendation reliability: the best solution depends on substrate, exposure, existing damage, and site conditions.

• Clear evidence: answers and reports should cite sources and avoid inventing product properties.

• Diagram correctness: generated diagrams must be labelled as explanatory visuals, not certified construction drawings.

• Deployment: a static website will need a secure backend API to run retrieval and AI generation.

6. Simple System Architecture

Website	→	RAG Backend	→	Knowledge Base
User question or request		Retrieves evidence and generates answers, reports, or diagrams		Technical documents and product data

7. Expected Result

The final product will be a website chatbot that answers waterproofing questions, suggests evidence-based solutions, generates simple technical diagrams, and prepares product comparison reports. It should show the sources used and indicate when a professional inspection or additional information is needed.

Note: The chatbot is a decision-support tool. It does not replace site inspection, manufacturer instructions, applicable standards, or professional engineering advice.